

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

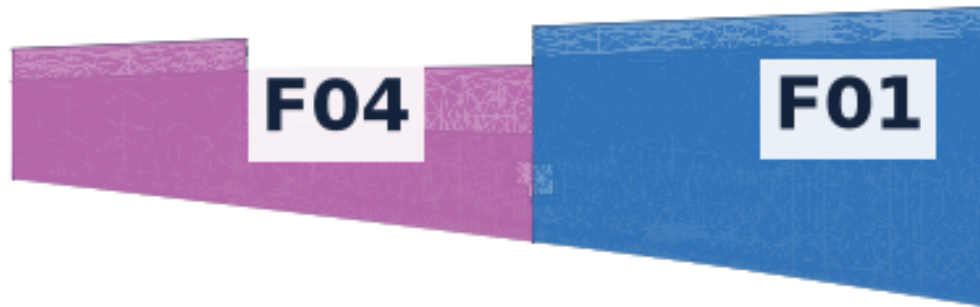
The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

Wings: match each skin to its side and span section

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

LEFT AIRCRAFT SIDE (-X)



RIGHT AIRCRAFT SIDE (+X)



 F04 Left outer wing A27

 F01 Left inner wing A52

 F02 Right inner wing A49

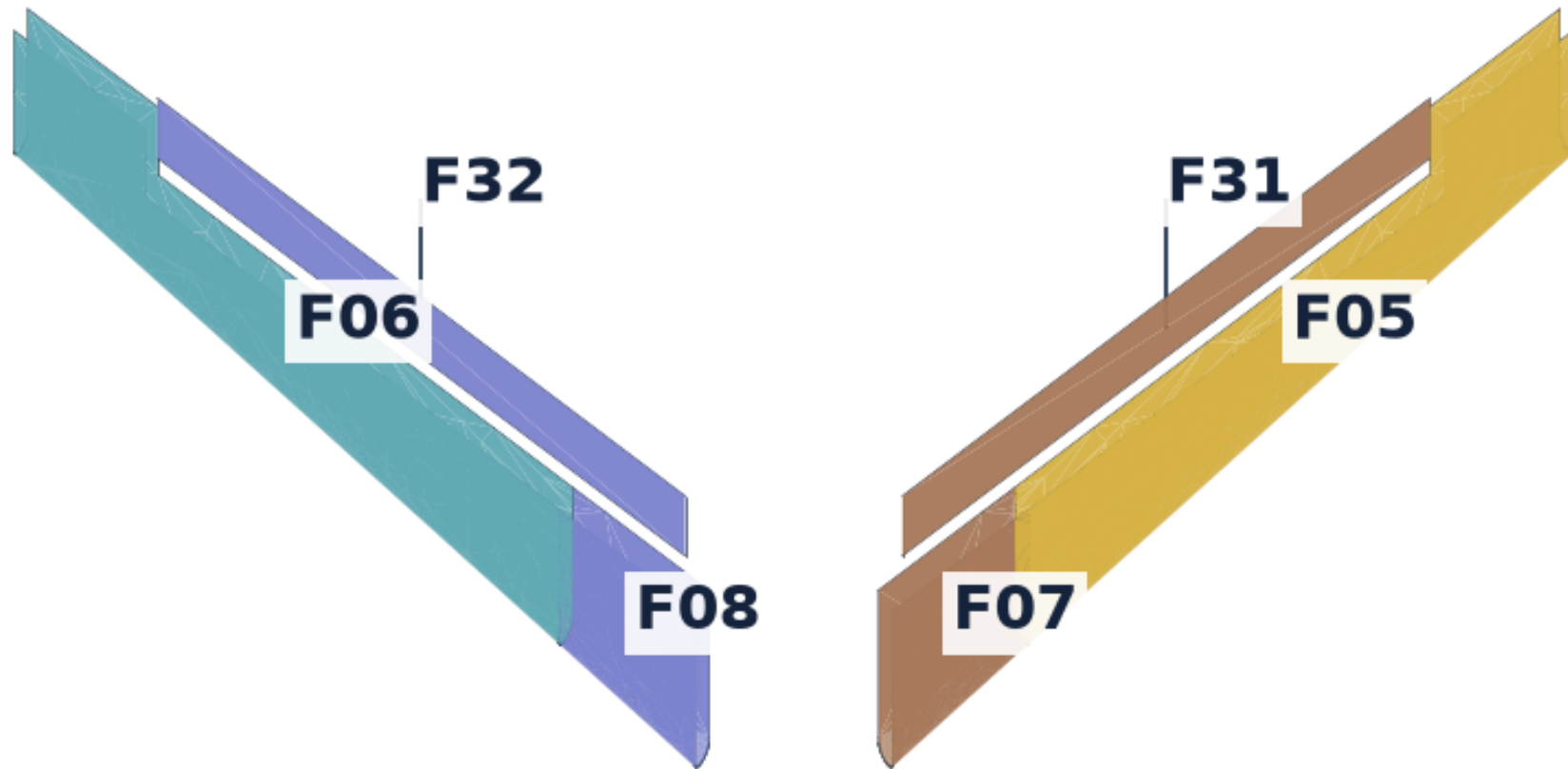
 F03 Right outer wing A38

Fit inner and outer sections to their ribs and spars before closing the trailing seams.

The narrow hinge-rail contact zones need inner-core relief; preserve the outer paper. See the forming guide.

V-tail: two fixed skins and one moving control on each side

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



- F06 Left main skin A22
- F08 Left root collar A22
- F32 Left control A60: cut two layers

- F05 Right main skin A57
- F07 Right root collar A57
- F31 Right control A41: cut two layers

Each fixed-tail pair meets at the root collar seam (80% of the CAD span parameter from the tip).

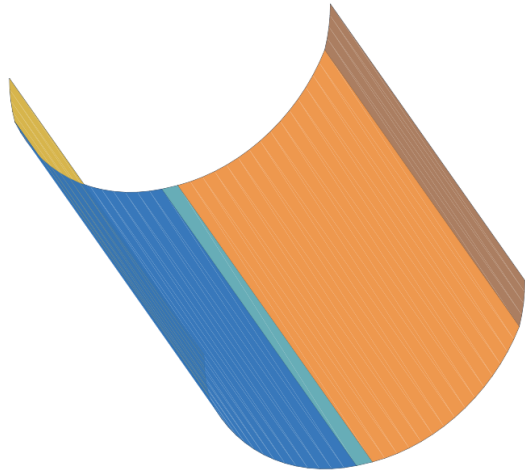
Control highlights show the CAD broad faces. Laminate two 5 mm blanks to 10 mm, then shape the R5 edges.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Lower fuselage: five distinct pieces around each section

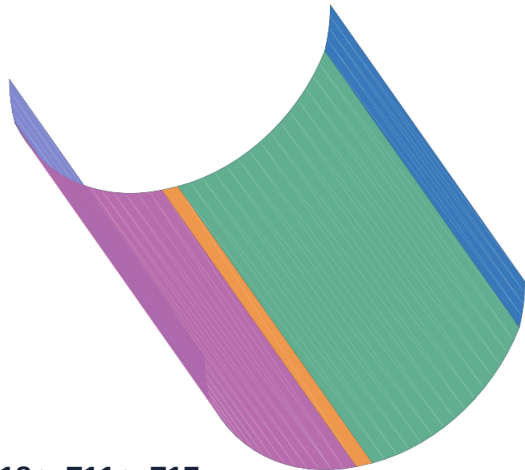
MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

FORWARD LOWER SHELL A56 | Y -335 to -130 mm



F13 > F09 > F14 > F10 > F15

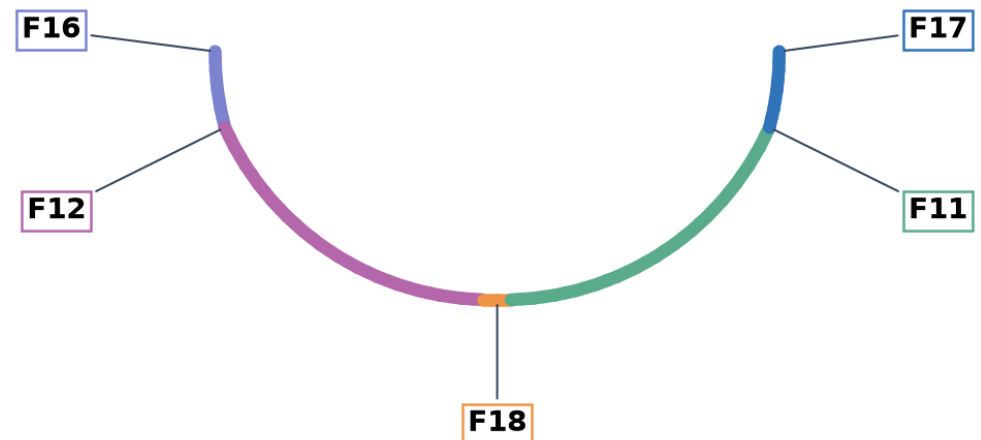
MIDDLE LOWER SHELL A17 | Y -130 to +238.65 mm



F16 > F12 > F18 > F11 > F17



Cross-section: left to right around bottom

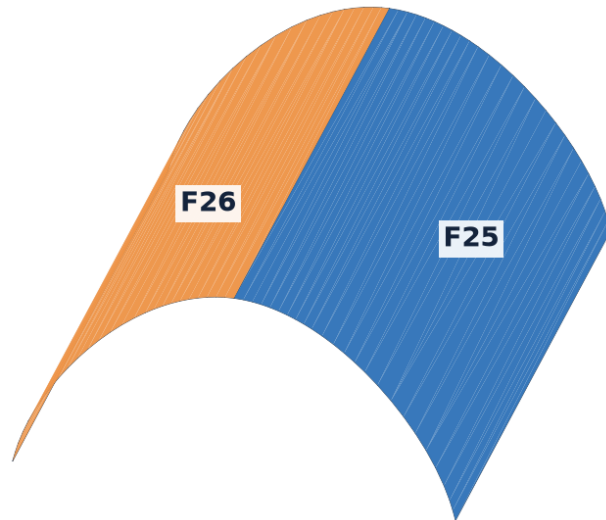


The narrow bottom strips are real parts; retain them.

Upper and aft-lower fuselage: match the coloured face pairs

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

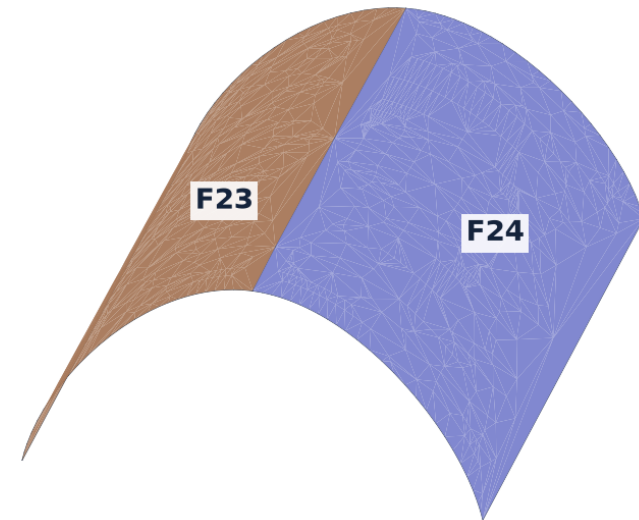
FORWARD UPPER A08 | Y -130 to 0



F26 left / F25 right; meet at the top.

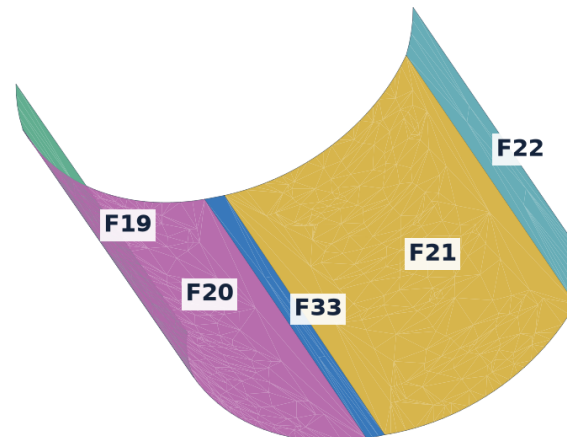
F23/F24 include a 1 mm paper-only lower-edge return: remove its foam, then trim the paper to fit.

AFT UPPER A15 | Y 233.65 to 389.65



F23 left / F24 right; meet at the top.

AFT LOWER A42 | Y 238.65 to 389.65 mm



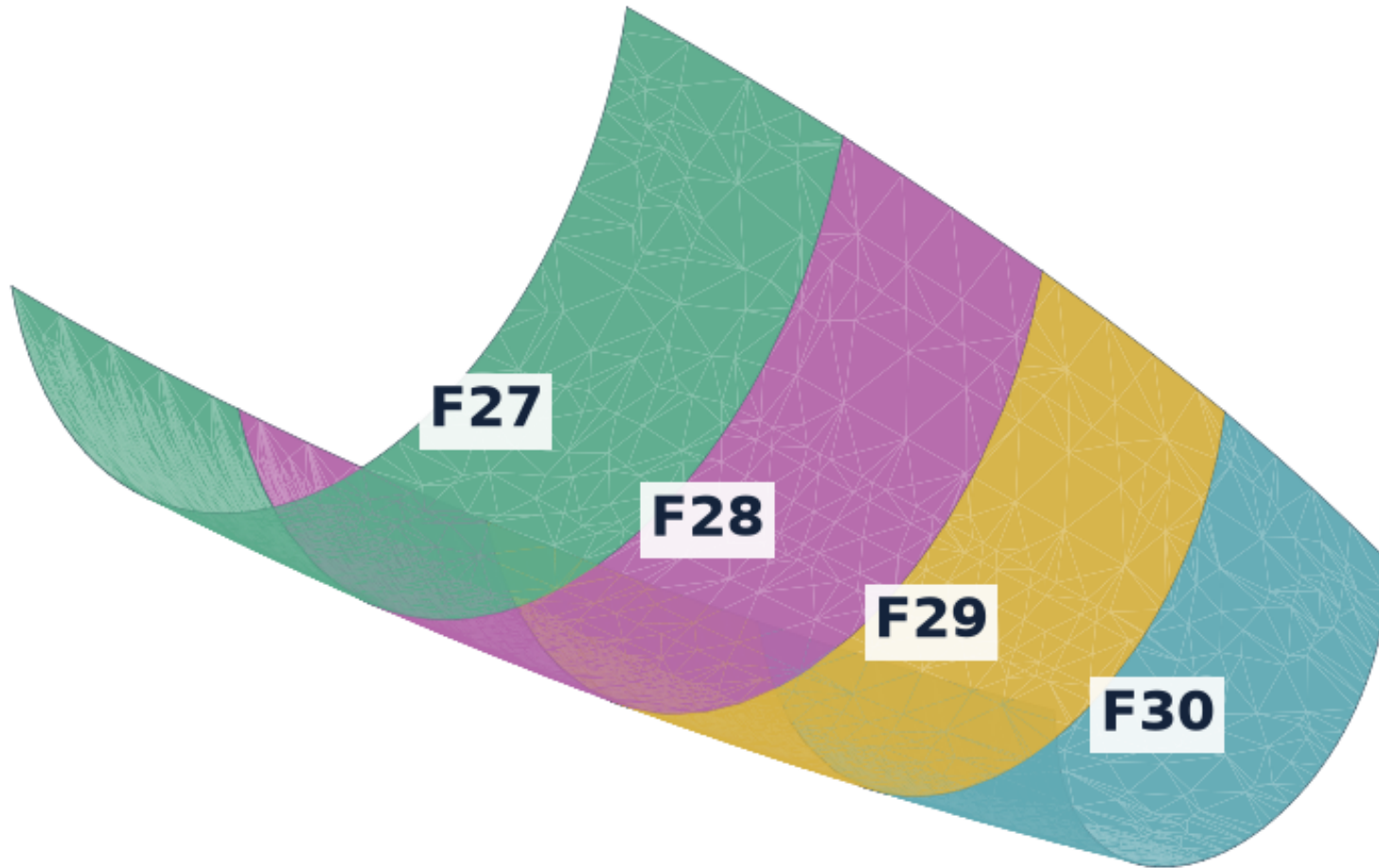
Around the underside: F19 left lip > F20 left curve > F33 bottom > F21 right curve > F22 right lip.

Fit the printed transitions between these shell groups before final skin bonding.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Rear underside: assemble four bands from front to aft

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



 F27 1 - forward band

 F28 2

 F29 3

 F30 4 - aft band

A20 assembly order: F27 > F28 > F29 > F30. The Y coordinate increases toward the tail.

Butt the outside-paper seam lines. Add separate inside straps; dry-fit around A43 and A62 before gluing.